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Comparing Public Acceptance of Dedicated CO₂ Storage and CO₂-EOR: Regional Trends and Their Impact on CCS Deployment

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Abstract

Public acceptance is recognised as a key factor in the success of carbon capture and storage (CCS) projects. However, societal views vary significantly depending on the storage method. This study evaluates regional trends and public discourses surrounding two principal CCS approaches: dedicated geological storage and CO₂-enhanced oil recovery (CO₂-EOR). By using surveys, focus groups, and case studies from North America, Europe, the Middle East, and Asia-Pacific, we identify common public concerns which include environmental risk, institutional trust, and questions of distributive fairness. In addition, we see region-specific viewpoints shaped by economic reliance on fossil fuels and different climate policy frameworks.

Dedicated geological CO₂ storage tends to be associated more directly with climate mitigation but often faces public doubts due to perceived local risks and limited economic benefits. On the other hand, CO₂-EOR is seen as more economically beneficial and is familiar in oil-producing areas, although it raises doubts about its validity as a climate solution. We investigate how these views affect permitting decisions, project timelines, and policy support. We argue that engaging with the public and incorporating their perspectives into project planning is crucial. By highlighting successful and unsuccessful cases, this study provides practical insights for developers and policymakers aiming to gain community support for CCS projects in various settings.

1. Introduction

The continuous release of greenhouse gases (GHGs) into the atmosphere is the main cause of global warming and climate change which leads to elevated mean global temperatures and disruptions across climatic, ecological, and societal systems (IPCC, [Masson-Delmotte et al., 2018](#)). The impacts of the global warming range from biodiversity loss to food insecurity and existential risks for vulnerable communities which reinforce the urgency of large-scale mitigation measures ([United Nations, 2023](#)). This urgency was highlighted in the Paris Agreement by urging nations to limit global temperature rise to well below 2°C, with goals to cap it at 1.5°C.

Carbon Capture, Utilisation, and Storage (CCUS) is one of the main pillars of mitigation strategies to reach the net-zero future (IPCC, 2018). The renewable energy technologies can displace many emissions sources. However, sectors such as heavy industry, aviation, and agriculture continue to produce residual emissions that are difficult to eliminate. Such gap can only be addressed by CCUS through capturing CO₂ at emission sources or directly from the atmosphere and storing it in deep geological formations (Metz et al., 2005).

One of the most commercially established applications of CCS is Enhanced Oil Recovery (EOR). In this process, CO₂ is injected into mature oil reservoirs to boost crude oil production while also storing CO₂ underground (Global CCS Institute, 2022). This technique has been applied for several decades to increase recovery factors by reducing oil viscosity and improving fluid mobility (Alvarado and Manrique, 2010). Due to the pressing environmental concerns of the global warming, the focus within CO₂-EOR has shifted toward maximising the volume of CO₂ stored per barrel of oil produced (Jessen et al., 2005; Eftehadtavakkol et al., 2014). However, the subsequent combustion of the recovered oil may counteracts part of the climate benefits of the stored CO₂, leading to growing recognition that CO₂-EOR represents a form of economically-beneficial utilisation rather than a form of net-negative carbon removal. Therefore, CO₂-EOR at best may be considered as a step along the path to net zero

In CCUS, public acceptance refers to the degree willingness of communities and society to allow, support, and live with the development and operation of CCUS projects (Upham et al., 2015). This acceptance is crucial for deploying CCUS projects. Attitudes vary significantly depending on the type of storage. Dedicated geological storage often faces skepticism because of the risks of long-term leakage and worries about continuing fossil fuel reliance (Ashworth et al., 2019). On the other hand, CO₂-EOR may gain more support in areas with strong oil and gas industries, as it is perceived to deliver both economic and environmental benefits. However, critics argue that it ultimately undermines climate goals by prolonging hydrocarbon extraction.

Public acceptance is also influenced by regional socio-political factors, regulatory transparency, and levels of community engagement. For instance, Scandinavian countries (e.g. Norway) show greater acceptance because of their inclusive policies and early involvement of stakeholders. In contrast, countries like Germany and Australia have seen public pushback with arguments that CCS can weaken climate policy and concerns about the safety of CCS projects (Otto, 2023). Additional factors, like tax credits and job benefits, can influence how communities view CCS, especially those that depend on enhanced oil recovery (Jenkins et al., 2017). Therefore, clear communication that emphasises local benefits and addresses risk concerns is vital for building public trust and addressing public concerns (Wolske et al., 2020).

This paper provides a structured comparison of how the public perceives dedicated CO₂ storage versus CO₂-enhanced oil recovery (CO₂-EOR), drawing on survey data, focus groups, literature review, and regional case studies. It maps the common concerns that cut across both approaches such as risk of leakage, fairness, and trust in institutions, while highlighting regional differences shaped by political context, reliance on fossil fuels, and climate policy frameworks. The analysis connects these perceptions to concrete project outcomes, including permitting delays, cancellations, and examples of successful deployment such as Shell Quest in Canada and Weyburn in Saskatchewan. By integrating insights from theory (social licence to operate, energy justice, risk perception) with evidence from real projects, the study distils lessons for developers and policymakers on how to incorporate public acceptance into CCS planning and decision-making, and how the framing of CO₂-EOR versus dedicated storage shapes its legitimacy as a climate solution.

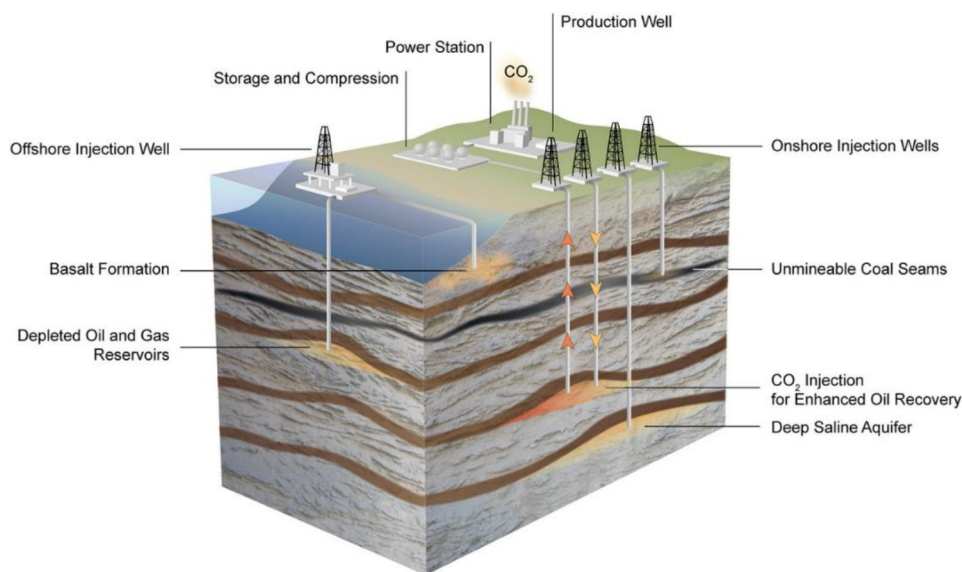


Figure 1—The different options of geological storage of CO₂ (Ali et al., 2022)

2. Background and Conceptual Framework

Public Acceptance of CCUS Projects

Public acceptance, including the local community, the regulator, the politicians, municipal government, and stakeholders, is a critical factor in the successful implementation of CCUS technologies. It reflects the level of support or opposition from society toward these projects. This support depends on perceived risks and benefits, trust in institutions, and fairness in decision-making processes (Huijts et al., 2007). Generally, public acceptance depends on three connected elements: the perceived balance between risks and benefits (e.g. climate mitigation vs. environmental hazards), the level of trust in project developers and regulatory authorities, and the procedural fairness of planning and consultation (Seigo et al., 2014; Jenkins et al., 2016). True public acceptance often involves active engagement, where stakeholders weigh the trade-offs between global climate benefits and local impacts (Wüstenhagen et al., 2007). CCUS projects, particularly those involving geological storage, require a strong level of societal legitimacy because public opposition can result in project delays, stricter regulatory requirements, or even cancellation. A notable example is the Barendrecht CCS project in the Netherlands, which was abandoned due to strong local opposition based on safety concerns and transparency issues (Feenstra et al., 2010).

Public acceptance operates at two distinct but connected levels: socio-political and techno-economic acceptability. The gap between these levels often explains why technically and economically sound projects face societal resistance.

- Socio-political acceptability is influenced by cultural norms, past experiences with extractive industries, and how the media presents them. For example, Norway's Sleipner CCS project gained from high public trust in Equinor and a clear national climate policy narrative (Tjernshaugen and Langhelle, 2009; Tjernshaugen et al., 2011). In contrast, Germany has largely rejected CCS, viewing it as "CO₂ dumping," which reflects a wider anti-fossil fuel sentiment (Lipponen et al., 2017; Otto, 2023).
- Techno-economic acceptability involves factors like cost, scalability, and following regulations. CO₂-EOR is often more appealing financially because of the oil revenues it generates. However, its socio-political acceptance is often challenged in areas with strong environmental activism or opposition to fossil fuels (Mabon et al., 2020).

When these two dimensions do not align, even technically sound and economically promising projects may encounter public rejection, especially if communities see them as unjust or not inclusive enough (Ashworth et al., 2015).

Theoretical Frameworks for Understanding Public Acceptance

Several conceptual frameworks have emerged to explain public responses to net-zero sustainable energy projects, offering insights into the social, psychological, and institutional factors at play (Fig. 2).

- **Social License to Operate (SLO):** It represents the degree of support or opposition that stakeholders—those who perceive their interests as being affected, whether positively or negatively—extend to a project, technology, or industry (Boutilier and Zdziarski, 2017; Gough and Mander, 2022). In this sense, SLO serves as a framework for acknowledging the broader social and political risks that can shape the success of a project or industry, moving beyond the limited view of community relations alone (Hall et al., 2015). Projects lacking SLO often encounter resistance, as seen in the Barendrecht and ZeroGen CCS projects in the Netherlands and Australia, respectively (Feenstra et al., 2010; Ashworth et al., 2012).
- **Energy justice concerns about fairness in energy projects (Taiwo and Tozer, 2025).** It focuses on three key aspects. First is distributive justice, which asks how the risks and benefits of a project are shared, who gains, and who bears the burdens. Second is procedural justice, which considers whether people have a fair chance to participate in the decisions that affect them. Third is recognitional justice, which is about acknowledging and respecting the values, identities, and needs of different groups. For example, in the case of CO₂-enhanced oil recovery (CO₂-EOR), questions of justice may arise if profits mainly go to private companies while the environmental risks are unfairly placed on already disadvantaged communities (Bomberg, 2017).
- **Risk Perception:** It refers to how people view and respond to potential dangers, which often differs from expert evaluations (Karimi and Komendantova, 2017). Research studies showed that the public tends to focus on "dread risks"—serious and uncertain threats like CO₂ leakage or earthquakes triggered by underground storage—especially when the technology is new or unfamiliar to them (Slovic, 1987; Tokushige et al., 2007). However, in regions where communities are already accustomed to oil and gas activities, CO₂-enhanced oil recovery (CO₂-EOR) may feel less novel and, as a result, less risky (Krause et al., 2014).
- **Trust and Transparency:** Public trust in institutions, including regulators and corporations, is a more reliable predictor of acceptance than technical assessments alone (Seigo et al., 2014). According to Faber and Sick (2022) In certain instances, institutions adopt measures that serve more as greenwashing tactics than genuine climate mitigation such as hidden trade-off, no proof, vagueness, worshipping false labels, irrelevance, lesser of two evils, and fibbing.

Conceptual Frameworks for Public Acceptance



Figure 2—Theoretical Frameworks for Understanding Public Acceptance

3. Public Awareness and Perception of CCS

Public awareness of CCS plays a crucial role in shaping how society views and accepts this technology. Recent public surveys conducted across 11 countries including Australia, Brazil, Canada, China, Denmark, France, Germany, Norway, the Netherlands, the United Kingdom, the United States, and Japan demonstrate consistently low levels of public familiarity with CCS technologies (Ha-Duong et al., 2009; Itaoka et al., 2009; Ashworth et al., 2013; Chen et al., 2015; Pianta et al., 2021; Merk et al., 2023; Andres et al., 2024; Toé et al., 2024). European countries generally report higher public awareness than China and the United States (Fig. 3). For instance, in Norway, known for its leadership in CCS development and frequent discussions on the topic, about 75% of respondents had heard of CCS and showed a good understanding of it (Merk et al., 2023; Anders et al., 2024). In other European countries and in Canada, awareness levels ranged from 46% to 54%. In contrast, 57% of U.S. respondents and 62% of Chinese respondents reported having never heard of CCS, respectively (Chen et al., 2015; Pianta et al., 2021). The differences in awareness levels may come from variations in media coverage, public involvement, and political discussion (Dowd et al., 2013). In the United States, for example, CCS gets little attention in both mass media and national policy discussions, which leads to lower public recognition (Dowd et al., 2013). Additional country-specific surveys further illustrate the variability in public awareness. In France, only 34% of respondents had heard of CCS (Ha-Duong et al., 2009), while in Australia, awareness was considerably higher at 77% (Ashworth et al., 2013). In Brazil, a 2024 study found that 55% of respondents had never heard of CCS (Toé et al., 2024). In Japan, the 2009 study by Itaoka et al. showed that 51% have heard of CCS and or know to some extent about it.

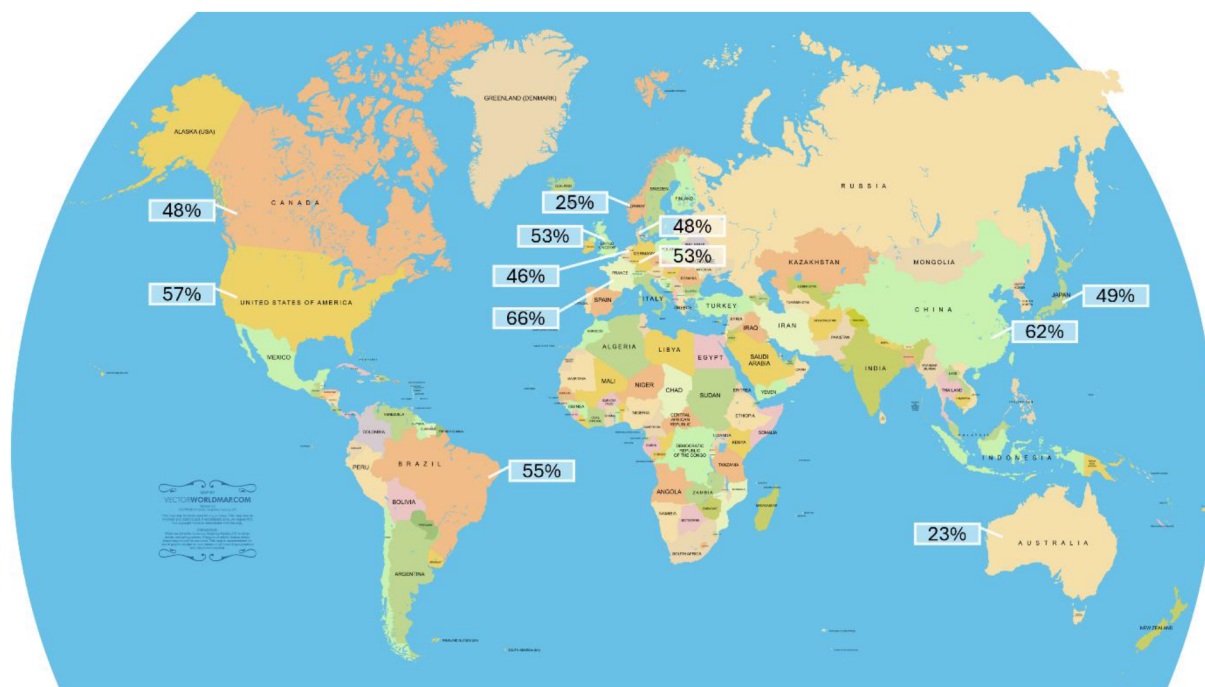


Figure 3—Results of surveys in 11 different countries about the knowledge of CCS. The map shows the percentages in the surveys that have never heard about CCS (Ha-Duong et al., 2009; Itaoka et al., 2009; Ashworth et al., 2013; Chen et al., 2015; Pianta et al., 2021; Merk et al., 2023; Andres et al., 2024; Toé et al., 2024).

The world map in Figure 3 shows absence of data in the area of the Middle East and Africa. While the region is recognised as a climate change hotspot with significant challenges, and there are discussions about CCS as a mitigation technology, direct studies on public awareness of CCS in North Africa and the Middle East are not as prevalent as in other global contexts. These strong global variations in the awareness of CCS emphasise the importance of targeted outreach and communication strategies to improve public understanding of CCS globally.

Surveys conducted in Canada, Germany, the Netherlands, and the UK by Anders et al. (2024) show that subjective knowledge, people's self-perceived familiarity with CCS, plays a key role in shaping public views on its implementation and related regulatory concerns. In the survey, people with little awareness often adopt neutral or uncertain positions when asked about the safety, necessity, or effectiveness of CCS. Levels of subjective knowledge also differ across demographics: the survey showed that younger people and men tend to report greater familiarity with climate mitigation technologies, including CCS, reflecting broader trends of interest and engagement with environmental issues (Anders et al., 2024).

Public Awareness and Perception of CO₂-EOR as a CCS Strategy

Public acceptance of CO₂-EOR as a CCS strategy varies significantly across regions due to differences in energy policies, economic reliance on fossil fuels, environmental advocacy, and media representation. This section synthesises previous studies and surveys to analyse public awareness and perception of CO₂-EOR in key regions.

United States

The U.S. has the most extensive CO₂-EOR operations globally, primarily in the Permian Basin and Gulf Coast regions. Public perception is generally more favourable than for dedicated CCS projects due to the economic benefits associated with increased oil production. Taking Wyoming as an example, a recent survey of Wyoming residents' values and beliefs related to energy in Wyoming was conducted by the University of Wyoming School of Energy Resources in fall 2022 (Western et al., 2023). The survey was distributed

to 2,800 randomly selected addresses in Wyoming, with 357 responses. Among respondents, 53.9% were between the ages of 18 and 50, while 46.1% were over 50. Gender representation was nearly equal, with females accounting for 48.9% of participants (Western et al., 2023). Respondents showed the highest levels of support for CO₂ capture associated with materials production and point-source emissions, indicating greater acceptance of technologies perceived as more familiar or directly linked to industrial processes and economic benefits (Fig. 4.a; Western et al., 2023). EOR also received moderate support (>50%; Fig. 4.a; Western et al., 2023). In contrast, results showed notable ambivalence and potential lack of knowledge toward direct air capture and permanent geologic storage, with many participants expressing neutrality or uncertainty (Fig. 4.a; Western et al., 2023). Notably, the most opposition was directed at the permanent geologic storage of CO₂ sourced from out of state, with 43% of respondents were opposing it (Fig. 4.a; Western et al., 2023). When asked directly about the EOR, almost half of the respondents were positive toward CO₂-EOR seeing that EOR is safe way to permanently store CO₂ while also producing oil. And it can contribute to the economy (Fig. 4.b; Western et al., 2023). This shows that most of respondents were directed toward the commercial benefit of CCS in opposition to the permanent geologic storage of CO₂.

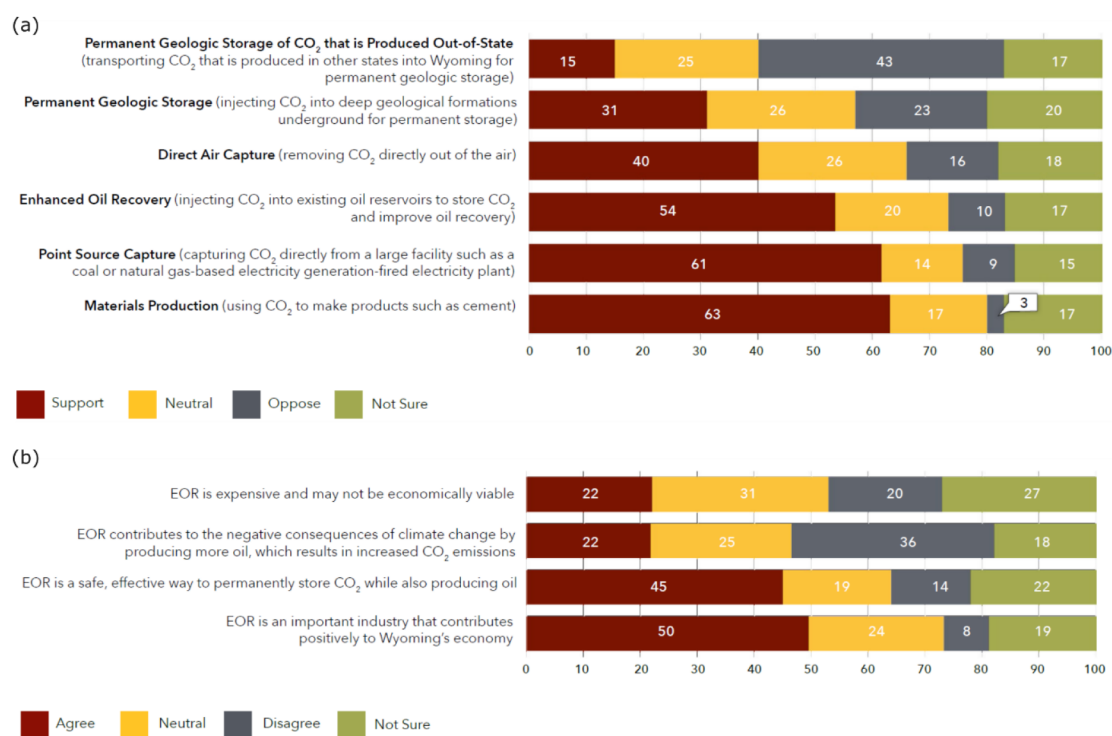


Figure 4—Survey outcomes of public Values and Preferences relating to Wyoming's Energy Future (Western et al., 2023). (a) Preferences regarding ways to capture and store CO₂. (b) Beliefs regarding the use of enhanced oil recovery for captured CO₂.

Canada

Similar to the USA, in Canada, public perception of CO₂-EOR has been shaped significantly by long-term community engagement, economic benefits, and trust in project operators. The outcomes of Boyd's (2015) study on Weyburn-Midale CO₂-EOR Project in Saskatchewan's public perception through interviews and community observations can serve as a good example. Residents of Weyburn and Goodwater, who have lived alongside the project for over a decade, generally expressed strong support because of its role in sustaining local employment and economic stability. Unlike broader carbon sequestration projects, which often face scepticism, the Weyburn initiative was perceived as part of the community rather than an outside environmental effort (Boyd, 2015). Economic benefits, including job creation and population growth, were highlighted as major advantages. In contrast, climate mitigation through CO₂ storage was rarely mentioned.

This aligns with studies that show local value alignment is crucial for public acceptance (Boyd, 2015; Western et al., 2023). Trust in local operators also mattered, as residents believed they could manage risks, even when concerns about CO₂ leaks arose (Boyd, 2015). Instead of resisting development, "not-in-my-backyard" (NIMBY) opposition, locals took pride in the project, seeing it as their own (Boyd, 2015). Generally, these findings may not generalise to other regions, as acceptance relies on context-specific factors such as economic dependency on fossil fuels and pre-existing trust in industry (Boyd, 2015). The Weyburn case highlights the importance of tailored communication strategies that focus on what matters most to communities over global just climate benefits which can increase the community support for CO₂-EOR initiatives.

Europe

In the North Sea region, public and stakeholder views on CO₂-EOR are influenced by the different narratives of energy transition, economic pragmatism, and climate change urgency (Mabon and Littlecott, 2015). Research from the UK by Mabon and Littlecott (2015) involved seven focus groups in Aberdeen, Edinburgh, and London. These groups included local residents, environmental organisations, energy experts, and investors, who discussed future scenarios for the North Sea. The aim was to understand whether CO₂-EOR is seen as a real part of a low-carbon transition or as a tool for maximising fossil fuel extraction (Mabon and Littlecott, 2015). The results showed that support for the technology depended on how the technology was framed: scenarios presenting CO₂-EOR as part of a managed decarbonisation pathway drew cautious approval, while framing it primarily as a way to maximise oil recovery was met with scepticism. Participants pointed out the importance of government leadership and clear policies to ensure that projects serve the public interest, indicating a lack of trust in market-driven approaches (Mabon and Littlecott, 2015). While fossil fuel dependence was acknowledged, some stakeholders pragmatically supported CO₂-EOR if it accelerated CCS deployment and reduced industrial emissions, though concerns about "greenwashing" persisted (Mabon and Littlecott, 2015). Overall, the findings suggest that CO₂-EOR in Europe will only gain a social licence if positioned as a bridge to CCS and long-term decarbonisation rather than as a tool to extend hydrocarbon extraction. Therefore, these results may indicate that European CO₂-EOR projects must focus on two key messages, highlighting short-term emissions reductions and long-term fossil fuel phaseouts, to balance economic needs with climate objectives (Mabon and Littlecott, 2015).

China

China considers CO₂-EOR a pivotal component of its broader CCUS strategy, especially given its heavy reliance on coal for energy and its' huge industrial emissions (Hill et al., 2020). therefore, CO₂-EOR is essential to China's decarbonisation efforts (Hill et al., 2020). Despite early adoption of CO₂ flooding, similar to the United States, our literature review shows a lack of surveys on public awareness and perceptions of CO₂-EOR in China. As the world's largest emitter of CO₂, this information is crucial for stakeholders, policymakers, and governments as they plan the future of CO₂-EOR and CCS in China.

Arabian Gulf Countries

CO₂-EOR is being actively explored in Gulf Cooperation Council (GCC) countries as a means to enhance hydrocarbon recovery while supporting carbon sequestration efforts. Saudi Arabia has pioneered regional CO₂-EOR applications, with the Uthmaniyah CO₂-EOR Demonstration project (operated by Saudi Aramco) successfully injecting captured CO₂ from the Hawiyah gas plant into the Ghawar field, demonstrating both incremental oil recovery and subsurface storage potential (Kokal et al., 2016). The UAE has made significant progress through ADNOC's Al Reyadah project, the world's first facility to capture CO₂ from the iron and steel industry. Each year, it captures around 800,000 tonnes of CO₂ from steel industries and supplies it for CO₂-EOR in the Bab and Rumaitha fields, making Al Reyadah a key driver of ADNOC's

2030 sustainability goals (ADNOC, 2020). However, broader deployment remains constrained by economic and logistical challenges, including high capture costs and limited CO₂ transportation infrastructure. Our literature review shows that public perception of CO₂-EOR in the Gulf is still understudied. At the same time, policymakers and industry leaders promote CCUS as part of a circular carbon economy Saudi plan to achieve net-zero emissions by 2060 (Saudi Green Initiative, 2021).

4. CO₂-EOR versus Dedicated Storage: Diverging Public Perceptions

Shared Public Concerns and Narratives

Public awareness of CCS technologies including both CO₂-EOR and dedicated CO₂ storage remains generally low worldwide (Tcvetkov et al., 2019). Surveys consistently show that many people have limited knowledge of what these projects entail, leading to high rates of "no opinion" responses and uncertainty in attitude polls (Rychlicki et al., 2015; IEA 2022). Despite this low familiarity, several common narratives and concerns emerge when the public is presented with the concept of injecting CO₂ underground.

One overarching public concern is environmental risk perception. Both CO₂-EOR and dedicated storage involve storing large volumes of CO₂ in geological formations, raising public questions about safety: Will the CO₂ remain contained, or could it leak and contaminate groundwater or air? Will injection trigger earthquakes or other unforeseen consequences? Such fears have been documented in multiple regions. For example, public opposition to onshore CO₂ storage in Europe has been driven by fears of leaks, induced seismicity, and the "industrialisation" of local landscapes (Tcvetkov et al., 2019). These perceptions of risk often far exceed expert estimates of actual risk, illustrating the classic gap identified in risk perception literature between technical risk and perceived risk. Dread factors (e.g. the notion of an invisible gas stored beneath one's community) and novelty play into this. The less familiar and more catastrophic a risk seems, the less acceptable it is to the lay public. Accordingly, an important part of managing public perception is risk communication and transparency. Studies have found that proactive engagement can significantly reduce perceived risks. For instance, the Aquistore project in Canada (a dedicated CO₂ storage demonstration) held open community open-houses with clear, accessible information on the project's purpose, injection depth, and monitoring plans; these efforts led to "extremely positive" local response and an observable reduction in community risk perceptions.

Another common narrative is the need for a "social license to operate." This concept, originating in social science research on extractive industries, refers to the informal acceptance and approval granted by the community and stakeholders, beyond mere legal permits. Both CO₂-EOR and dedicated storage projects must cultivate this social license by aligning with community values and involving the public in a meaningful way. Experience shows that if local people feel they have no voice or stake in a project, opposition can mount even if objective risks are low (Tcvetkov et al., 2019). Conversely, when community members see that their concerns are heard and addressed – for example, through public consultations, community benefits, or participatory monitoring – they are more likely to consent to the project. This was evident in the case of Saskatchewan's Aquistore, where an outreach advisory committee and regular updates to local landowners helped secure broad support (Sacuta, et al., 2017).

Different Narratives and Perceived Purpose

Despite these similarities, the public narratives diverge in important ways when it comes to CO₂-EOR versus dedicated CO₂ storage, largely due to differences in perceived purpose and climate framing. Dedicated CO₂ storage is typically framed as a climate change mitigation strategy, CO₂-EOR, on the other hand, has an inherently dual purpose: it stores CO₂ while extracting additional oil. This duality gives rise to more complex public interpretations. On one side, industry proponents and some policymakers frame CO₂-EOR as a pragmatic "win-win" approach: it can drive industrial and economic benefits (by boosting oil production

and revenue) that in turn make CCS financially viable, all while storing CO₂ that would otherwise be emitted (Alvarado, et al., 2010). In this positive narrative, CO₂-EOR is portrayed as a bridge to a lower-carbon future – a way to kick-start carbon storage efforts by leveraging oil revenues and demonstrating CO₂ injection at scale (Alvarado, et al., 2010). Some members of the public, especially in regions with oil-dependent economies, do resonate with this framing. Surveys in petroleum-producing regions find a significant share of residents' view CCS and CO₂-EOR as opportunities for job creation and economic development, and thus evaluate them positively (e.g. Boyd, 2015; Western et al., 2023).

On the other side of the narrative, many observers – particularly environmental NGOs, climate conscious citizens, and even neutral laypersons when the facts are explained – express doubts or opposition to CO₂-EOR on climate grounds. The fundamental critique is that, by facilitating additional oil extraction and eventual combustion of that oil, CO₂-EOR might undermine its own climate benefit. In the words of one stakeholder, "it contributes to the negative consequences of climate change by producing more oil". Indeed, public focus groups in the UK showed that scenarios "emphasising maximising oil recovery" with CO₂-EOR were met with marked cynicism, or even outright opposition. Many participants perceived such plans as oil companies "dressing up" an oil project as a climate solution, fuelling worries of greenwashing (Mabon, et al., 2016). This aligns with findings in the literature that the climate framing of CO₂-EOR is fraught: while technically it involves CO₂ storage, people do not automatically equate it with a climate solution because the intent (oil production) appears at odds with climate goals.

By contrast, dedicated CO₂ storage is more straightforwardly associated with climate mitigation – there is no additional fossil fuel output – which can make it more acceptable to environmental stakeholders. For instance, environmental groups in Norway have largely supported dedicated CCS projects as a key method to combat global warming, whereas in Germany environmental groups have tended to oppose CCS (and especially EOR) due to perceived risks and the sense that it prolongs reliance on fossil energy (Tjernshaugen and Langhelle, 2009; Tjernshaugen et al., 2011; Lipponen et al., 2017; Otto, 2023).

Finally, CO₂-EOR tends to be publicly debated along the lines of climate compromise versus economic pragmatism, dedicated CO₂ storage is framed as a climate necessity weighed against local impacts. Each requires a tailored narrative strategy. Where they converge is in the need for trust: if either type of project is perceived as a "backroom deal" benefiting only its proponents, public acceptance will likely falter. If instead the project is framed as serving the public interest, with evidence to back that up, then even those initially skeptical can become more supportive (Tcvetkov, et al., 2019; Mabon, et al., 2016).

5. Implications for CCS Deployment

Integrating Public Acceptance into CCS Project Planning

Public acceptance is widely recognised as a critical factor that can make or break Carbon Capture and Storage (CCS) projects. Integrating public acceptance into project planning means moving beyond a mere "checkbox" approach and proactively embedding stakeholder engagement, transparency, and responsiveness throughout the project lifecycle. Researchers emphasise that gaining a "social license to operate" is as vital as obtaining regulatory permits (Terwel, et al., 2011; De Best-Waldhober and Daamen, 2006). In practice, this entails early and continuous dialogue with stakeholders, incorporation of community feedback into project design, and clear communication about both benefits and risks. For example, social site characterisation frameworks have been developed to help CCS planners understand local contexts (values, concerns, history) before finalising project decisions. This aligns with social science theories such as stakeholder theory, which calls for identifying and addressing the interests of all affected groups (local residents, indigenous communities, NGOs, etc.) in project decision-making.

Risk perception and trust are particularly important to account for in planning. Studies show that public perceptions of CCS risks are not purely technical judgments but are deeply influenced by emotional and

cognitive factors – for instance, fears of CO₂ leakage or induced seismicity may loom large even if experts assess risks as low. Effective planning therefore involves transparent risk communication and third-party validation to build trust.

Public Response Case Studies: Delays, Cancellations and Support

Public responses to CCS projects globally have ranged from staunch opposition that halted projects, to supportive community partnerships that enabled project success. Historically significant cases in Europe illustrated early on how public opposition could derail CCS. A notorious example is the Barendrecht project in the Netherlands. This onshore storage proposal (to inject CO₂ beneath a suburban town) faced intense local opposition in 2009–2010, with residents and the municipal council revolting against being a "CO₂ backyard." Despite initial government approval and funding, the project was first delayed and ultimately cancelled in late 2010 due to public outcry. The community saw no local benefits but plenty of perceived risks, and neither Shell nor the national authorities engaged the public adequately until opposition had already hardened (Herzog, 2017).

Public response has not been universally negative, however. There are also cases of successful public engagement and support that offer a counterpoint. In North America, the Shell Quest project in Alberta, Canada provides a notable example of a CCS project that achieved and maintained public acceptance. Quest, a large-scale CCS facility capturing CO₂ from oil sands operations, was launched in 2015 and proceeded with relatively smooth community relations. Shell applied lessons learned from Barendrecht – implementing an "excellent outreach program" for Quest that involved extensive community consultation, town-hall meetings, and even informal coffee chats to build familiarity and trust. This proactive engagement helped Quest secure a social license to operate; the local community and First Nations were largely supportive or neutral, perceiving the project's climate benefits and safety measures as credible. Independent reviews have pointed to Quest as a best practice model for CCS outreach, noting that early transparency and local inclusion helped avoid the opposition seen elsewhere (Herzog, 2017).

Overall, these cases highlight a duality of challenges and successes. On one hand, inadequate handling of public concerns has led to delays or cancellations of several high-profile CCS projects. Opposition often arises when communities feel risks are imposed on them for others' gain, or when they distrust the project proponents' motives. Social science frameworks like the "social gap" or social license concept underscore that beyond formal permits, projects need broad societal consent to proceed. Public opposition has even influenced policy. On the other hand, the success stories demonstrate that public support can be earned. When developers actively engage communities with respect, transparency and responsiveness, it is possible to secure at least grudging acceptance, if not enthusiastic support. Indeed, several CCS projects have managed to obtain a "social license to operate" by building trust: they approached engagement as a dialogue, acknowledged wider community values (like climate benefits and local development), and addressed issues of procedural justice (fair process) and distributional justice (fair share of benefits/risks).

6. Conclusions

This study highlights the nuanced and context-dependent nature of public acceptance for CO₂-EOR and dedicated CO₂ storage. While both technologies involve injecting CO₂ into subsurface formations, they are perceived through distinct social and political lenses. Dedicated storage is typically viewed as a climate solution but can face opposition when local communities perceive high environmental risks or feel excluded from decision-making. CO₂-EOR, meanwhile, benefits from its economic framing and operational familiarity in hydrocarbon-producing regions, but its association with continued fossil fuel extraction undermines its credibility in climate discourse.

Key findings include:

- Public acceptance is not a fixed barrier but a dynamic outcome of engagement, framing, and trust. Early, transparent, and participatory project design significantly increases the likelihood of social support.
- Successful cases, such as Shell's Quest project in Alberta and Aquistore in Saskatchewan, demonstrate the value of sustained community dialogue, responsive risk communication, and visible local benefits.
- Conversely, the failure of the Barendrecht project in the Netherlands illustrates the consequences of delayed engagement and perceived procedural injustice.

Integrating public acceptance into CCS deployment is not just a communications challenge—it is a strategic imperative. Project developers and policymakers must move beyond informational outreach and toward collaborative planning models that address local concerns and climate imperatives in tandem. Future research should further investigate regional case studies, cross-cultural dynamics in trust formation, and the long-term evolution of public opinion as CCS technologies mature.

7. References

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